Smart Personalized Product Suggestion System

Project Workflow, Architecture & Proposal

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Delivery Timeline	3 Weeks
Version	v1.0

1. Executive Summary

This document outlines the complete workflow, technical architecture, feature set, and project roadmap for the Smart Personalized Product Suggestion System — an AI-powered solution designed to help customers choose the right products faster through a conversational questionnaire experience.

The system will be deployed across three channels — WhatsApp, Instagram, and a web landing page — and will deliver three tiers of personalized product offers (Basic, Intermediate, and Premium) tailored to each customer's responses.

Core Objective

Combine speed and personalization to simplify the customer decision-making process, increase conversion rates, and improve overall customer experience — fully deployed and live within 3 weeks.

2. Project Scope

2.1 What Is Included

- Design of customer questionnaire (5–8 smart, conversion-focused questions)
- AI-powered recommendation engine built with Python and LangChain
- Product catalog structuring and integration
- WhatsApp Business API setup and integration
- Instagram Messenger API setup and integration
- Web landing page with embedded questionnaire
- 3-tier personalized offer system (Basic, Intermediate, Premium)
- Portuguese language support with English-ready architecture
- Full testing across all channels before launch
- Handover documentation and walkthrough session
- Post-launch support period

2.2 What Is Not Included (Base Scope)

- CRM integration (available as an add-on in Premium tier)
- Paid advertising campaigns or social media management
- Ongoing maintenance beyond the support period
- Custom mobile app development

3. How the System Works

The system follows a simple 4-step flow for every customer interaction:

Step	Stage	What Happens	Customer Experience
01	Customer Starts	Customer reaches out via WhatsApp, Instagram DM, or visits the web page	Simple, familiar channels — no app to download
02	Questionnaire	Customer answers 5–8 short personalized questions about their needs, preferences, and budget	Fast, conversational — takes under 2 minutes
03	AI Processing	LangChain AI engine analyzes answers and matches customer to the best products from the catalog	Instant — no waiting, no human needed
04	Offer Delivery	Customer receives 3 personalized offer options: Basic, Intermediate, and Premium	Clear choices — easy decision, higher conversion

4. Questionnaire Design

The questionnaire is designed for maximum completion rate. Questions are short, conversational, and focused on capturing the three most important buying signals: need, preference, and budget.

4.1 Sample Question Framework

- What problem are you trying to solve? (Multiple choice — 4 options)
- Who is this product for? (You / Someone else / A business)
- What is most important to you? (Price / Quality / Speed / Features)
- Do you have a preferred style or type? (Category-specific options)
- What is your budget range? (4 budget tiers)
- How urgently do you need this? (Today / This week / Just browsing)
- Any specific requirements or preferences? (Short open text — optional)

Note
The final questionnaire questions will be refined after reviewing your specific product catalog to ensure maximum relevance and conversion effectiveness.

5. Technical Architecture & Stack

The system is built on modern, scalable technologies chosen for reliability, speed, and ease of future expansion.

Component	Technology	Purpose
Recommendation Engine	Python + LangChain	AI-powered product matching
Language Model	Claude API (Anthropic)	Natural language understanding & personalization
WhatsApp Channel	Twilio WhatsApp API	Customer interaction via WhatsApp
Instagram Channel	Meta Messenger API	Customer interaction via Instagram DM
Web Channel	Streamlit / HTML	Embedded questionnaire on web/landing page
Product Catalog	JSON / SQLite	Structured product data storage
Hosting	Cloud (AWS / Railway)	Reliable, scalable deployment
Language	Portuguese (primary)	EN-ready architecture for future expansion

5.1 Architecture Overview

- Customer sends a message via WhatsApp / Instagram / Web
- Message is received by the backend server hosted on the cloud
- LangChain AI engine processes the questionnaire responses
- Engine queries the product catalog and generates 3 personalized offers
- Offer is sent back to the customer through the same channel
- All interactions are logged for performance monitoring

6. Service Tiers & Feature Comparison

Three service tiers are available to match different business needs and budgets. All tiers include the core recommendation system — higher tiers add more channels, capacity, and support.

Feature	Basic	Intermediate	Premium
Questionnaire Design	5 questions	7 questions	8+ questions
Recommendation Engine	Rule-based logic	AI-powered (LangChain)	AI-powered + learning
Product Catalog Setup	Up to 20 products	Up to 50 products	Unlimited products
WhatsApp Integration	✓ Basic setup	✓ Full integration	✓ Full + automation
Instagram Integration	✗ Not included	✓ Basic setup	✓ Full integration
Web Landing Page	Simple form	Branded page	Custom branded page
Offer Tiers	3-tier system	3-tier system	3-tier + upsell logic
Language Support	Portuguese only	PT + EN ready	PT + EN + scalable
CRM Integration	✗ Not included	✗ Not included	✓ Optional add-on
Documentation	Basic guide	Full guide	Full guide + video
Post-Launch Support	3 days	7 days	14 days
Delivery Timeline	2 weeks	3 weeks	3–4 weeks

6.1 Recommended Tier

Recommendation: Intermediate Tier
Based on your goals — full WhatsApp + Instagram integration, AI-powered personalization, Portuguese-first with English expansion capability, and a fully deployed live system — the Intermediate Tier covers everything you need at the right scope and timeline.

7. Project Roadmap — 3-Week Delivery

The project is divided into 6 phases designed for fast delivery without sacrificing quality. Each phase has a clear output so you can track progress at every step.

Phase	Timeline	Tasks	Deliverable

Phase 1	Days 1–3	Kickoff call, review product catalog, finalize questionnaire structure, confirm channel access	Approved questionnaire + catalog structure
Phase 2	Days 4–8	Build AI recommendation engine, integrate LangChain + product catalog, test 3-tier offer logic	Working recommendation engine
Phase 3	Days 9–14	WhatsApp Business API setup, Instagram Meta API setup, connect recommendation engine to both channels	Live multi-channel integration
Phase 4	Days 15–18	Build web landing page with embedded questionnaire, connect to recommendation engine	Live web channel
Phase 5	Days 19–21	Full QA testing across all channels, bug fixes, Portuguese language review, performance check	Tested system ready for launch
Phase 6	Day 21	Deploy to production, handover documentation, walkthrough call, post-launch support begins	Fully deployed live system

8. What I Need From You

To get started immediately, I will need the following from you:

Item	Details
Product Catalog	Export of your products (Excel, CSV, or any format you have)
Channel Access	Admin access to WhatsApp Business and Instagram Business accounts
Branding	Logo, brand colors, and any design preferences (if available)
Kickoff Call	30-minute call to align on questionnaire design and priorities
Feedback Availability	Availability to review and approve at end of each phase

9. Next Steps

Once you review this document, here is how we move forward:

- Review this proposal and confirm the service tier
- Agree on final pricing and timeline
- Place the order on Fiverr to officially start the project

- Schedule a 30-minute kickoff call
- Share your product catalog and channel access
- Development begins — Phase 1 starts within 24 hours of kickoff

Ready to Move Fast?
I've structured this project to deliver a fully working system in 3 weeks. Once you confirm, we can start immediately. I'm looking forward to building this with you.

Prepared by Mustafa